

37495 Statistical Design and Models for Evaluation Studies

Computer Lab 1

Problem A

- a) Calculate $\int_0^3 \cos(x)e^{-x^2} dx$.
- b) Determine $\min\{\sin(n) : n = 0, \dots, 1000\}$. Where is this minimum attained?
- c) Determine the minimum of the function $f : (0, \infty) \rightarrow \mathbb{R}_+$, $x \mapsto 1/x + e^x$.
- d) Plot the function $f(x) = 1 + x + x^2 + x^3 + x^4$ on the interval $[-1, 1]$.
- e) Plot all roots of $f(x) = 1 + x + x^2 + x^3 + x^4$ in the complex plane (use `polyroot()`).

Problem B

- a) Calculate the Gram matrix $G = X^\top X$ of the column vectors from the matrix

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 9 \\ 1 & 8 & 0 \end{bmatrix}.$$

- b) Determine eigenvalues and eigenvectors of G .
- c) Determine $G^{1/2}$ and $\exp(-G)$ (use `expm` package).

Problem C

- a) Build a sequence $(x)_j$ ranging 0.3 to 9.1 with step size 0.1. Determine its length n .
- b) Obtain the sequence $(y_j = x_j^2 - x_j - 10)_{j=1}^n$. Determine which values appear in y twice or more (use `uplicated()`).
- c) Obtain a sequence z by omitting the duplicated values.
- d) Order z in a decreasing order.

- e Drop all entries in z which belong to even indices.

Problem D

Set the random number generator in a pre-determined state by `set.seed(1234)`.

- a) Build a random sequence `x <- rnorm(60)` of length 60. Fill two 4×15 matrices Y and Z by the numbers from x , Y row by row and Z column by column.
- b) Determine the mean for each row and for each column of Z (use `apply`).
- c) Create three-dimensional array X by `X <- x` and then `dim(X) <- c(3, 4, 5)` and calculate a matrix $M = (M_{ij})_{i=1, j=1}^{3, 4}$ given by $M_{i,j} = \sum_{k=1}^5 X_{i,j,k}$ (use `MARGIN` in `apply`).